Multimedia Tools and Applications (Springer Nature)

Original Research Article • October 2026

RESEARCH ARTICLE — IMAGE RESTORATION & MULTIMEDIA SYSTEMS

NeuroClean: High-Fidelity Smartphone Image Denoising via Hybrid U-Net–DnCNN Dual-Stream Deep Learning and Overlapping Hann-Window Tiling

Majji Naveen[1], **Parri Hari Chandan**[1], **M. Siva Charan Reddy**[1], **and Jayalakshmi M**[1*]

[1] School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology (VIT), Vellore, Tamil Nadu 632014, India

Author Emails: majji.naveen2023@gmail.com (M. Naveen), harichandan.parri2023@vitstudent.ac.in (P. H. Chandan), mareddy.siva2023@vitstudent.ac.in (M. S. C. Reddy)

Supervised by & Corresponding Author: Dr. Jayalakshmi M (Associate Professor, SCOPE, VIT Vellore, Email: m.jayalakshmi@vit.ac.in)

Abstract

Smartphone cameras operate under acute physical form-factor limitations: micro-apertures and sub-micron to micron pixel pitches (1.0 µm – 1.4 µm) restrict photon capture, yielding severe Poisson photon shot noise, amplifier read noise, and chromatic grain under low-ambient illumination. Existing neural restoration approaches either suffer from oversmoothing of high-frequency photographic textures (in deep multi-scale autoencoders like U-Net) or fail to preserve low-frequency luminance continuity across large receptive fields (in feed-forward residual networks like DnCNN). Furthermore, deploying deep models on arbitrary, high-resolution smartphone captures (12+ to 100+ megapixels) induces fatal memory exhaustion or catastrophic boundary grid artifacts under naive spatial tiling. In this paper, we present **NeuroClean**, a unified deep learning and production deployment framework for smartphone image restoration. NeuroClean introduces a dual-stream **Hybrid U-Net–DnCNN** neural architecture that synergizes: (i) a multi-scale contracting-expansive U-Net branch with long skip connections to capture scene context and uniform illumination gradients, and (ii) a non-pooling 10-layer residual DnCNN branch operating at full spatial resolution to isolate high-frequency sensor noise without spatial decimation. The feature representations from both streams are fused through a 1×1 cross-channel projection layer to estimate the underlying residual noise field R(y), yielding the restored image via direct residual subtraction $\hat{x}$ = clip(y - R_hybrid(y), 0, 1). To process arbitrary image dimensions without memory bloat or edge stitching seams, we engineer a **2D Hann-window overlapping cosine tiling engine** with reflect padding and weighted overlap-add reconstruction. Evaluated on paired real-world smartphone photographs (1404×2040 native resolution), our hybrid model achieves a mean Peak Signal-to-Noise Ratio (PSNR) of **28.60+ dB** and Structural Similarity Index (SSIM) of **0.8920+**, representing a +4.56 dB and +0.1915 improvement over the raw degraded baseline (24.04 dB, 0.7005 SSIM), outperforming standard DnCNN (27.42 dB, 0.8650) and 3-level U-Net (27.95 dB, 0.8710) with statistically significant gains under Holm-corrected Wilcoxon signed-rank tests ($p < 0.001$). The complete system is deployed as an open-source, production-grade cloud application featuring an asynchronous FastAPI inference engine, a responsive glassmorphic frontend with interactive comparison sliders, real-time blind noise estimation (σ̂), and an interactive 3D WebGL photodiode sensor simulation.

Keywords: Smartphone image denoising · Computational photography · Deep residual learning · U-Net · DnCNN · Hann-window tiling · Photon shot noise · Multimedia systems.

1 Introduction

Computational photography has emerged as the definitive driver of image quality improvements in modern mobile consumer electronics. Over the past decade, smartphones have superseded dedicated digital single-lens reflex (DSLR) and

mirrorless cameras for everyday photographic capture [1]. However, smartphone camera hardware remains constrained by severe physical limitations. While full-frame and APS-C camera sensors possess pixel pitches between 3.7 μm and 8.4 μm, mobile camera sensors must compress tens of millions of photosensitive diodes onto sensor dies smaller than 1/1.3 inches, yielding pixel pitches typically spanning 0.8 μm to 1.4 μm [2].

Under low-light ambient conditions, high shutter speeds, or high ISO settings, the reduced physical collection area of each photodiode severely restricts the number of incident photons [3]. The resulting captures are degraded by composite stochastic noise processes: (1) *Photon shot noise*, arising from the Poisson distribution of photon arrival times [4]; (2) *Thermal and analog read noise*, generated by charge-to-voltage conversion amplifiers during analog-to-digital conversion (ADC) [5]; and (3) *Chromatic noise*, resulting from spatial interpolation during Bayer color filter array (CFA) demosaicing [6].

Historically, image denoising relied upon handcrafted spatial and transform-domain filters, such as Bilateral Filtering [7], Non-Local Means (NLM) [8], and Block-Matching and 3D Filtering (BM3D) [9]. While mathematically elegant, these techniques assume additive white Gaussian noise (AWGN) with stationary variance across the entire image plane. In real smartphone image signal processors (ISPs), however, noise is signal-dependent, heteroscedastic, and spatially correlated across RGB color channels [10]. Consequently, classical filters frequently blur fine organic textures (e.g., foliage, facial hair, fabric weaves) while leaving low-frequency chroma splotches intact.

The advent of deep convolutional neural networks (CNNs) revolutionized image restoration. Zhang et al. introduced DnCNN [11], demonstrating that predicting the residual noise field rather than the clean image directly accelerates training convergence and boosts restoration fidelity. Concurrently, Ronneberger et al.'s U-Net [12] gained widespread adoption in image-to-image translation tasks due to its multi-scale contracting-expansive paths and long skip connections that preserve macro-structural context [13].

Despite these advances, three critical bottlenecks persist: (1) *The Context vs. Grain Dilemma*: DnCNN maintains native resolution and isolates fine pixel grain, but its receptive field is bounded by network depth (21×21 pixels for 10 layers), rendering it blind to wide-area luminance mottling. Conversely, U-Net captures broad context via pooling, but destroys fine textures and introduces spatial phase distortions upon upsampling. (2) *The High-Resolution Boundary Artifact Barrier*: Direct evaluation on 12+ megapixel images triggers Out-Of-Memory (OOM) GPU exceptions, whereas naive non-overlapping patch tiling introduces unacceptable orthogonal boundary seam lines. (3) *The Prototype-to-Deployment Disconnect*: Research frequently terminates with command-line notebooks on downsampled crops rather than responsive, privacy-preserving, production cloud web applications supporting arbitrary dimensions.

1.1 Research Contributions

1. **Dual-Stream Hybrid Architecture:** A novel Hybrid U-Net–DnCNN deep neural network combining multi-scale contextual feature extraction (Stream 1) with non-pooling high-frequency residual grain modeling (Stream 2) joined via a 1×1 cross-channel fusion projection layer.
2. **Deep Residual Formulation:** Mathematical formulation of the restoration objective as deep residual subtraction $\hat{x} = \text{clip}(y - R_{hybrid}(y), 0, 1)$, demonstrating faster convergence and superior preservation of image structural fidelity.
3. **2D Hann-Window Overlapping Cosine Tiling Engine:** An engineered patch-inference engine utilizing 2D separable Hann cosine windows with 50% overlap and reflection padding, establishing a mathematical partition of unity that guarantees zero boundary seams across arbitrary input resolutions.
4. **Comprehensive Empirical Benchmarking:** Rigorous quantitative evaluation across paired real-world smartphone test sets (1404×2040 native resolution), achieving 28.60+ dB PSNR and 0.8920+ SSIM, significantly outperforming comparative baselines with Holm-corrected Wilcoxon signed-rank significance ($p < 0.001$).
5. **Full-Stack Production Deployment & 3D WebGL Simulation:** An open-source, full-stack cloud application featuring an asynchronous FastAPI microservice, a responsive React/TypeScript frontend with glassmorphic UI/UX, Dark/Light theme switching, live metrics, and an interactive 3D Three.js photodiode sensor simulation.

2 Related Work and Literature Review

2.1 *Smartphone Image Noise Physics:* Smartphone sensors operate with pixel pitches between 1.0 μm and 1.4 μm [2], [14]. Under photon-starved captures, quantum fluctuations in photon arrival follow a Poisson distribution: P(k; λ) = (λ^k e^(-λ)) / k! [4]. Combined with thermal agitation and readout fluctuations modeled as zero-mean Gaussian noise N(0, σ_r^2), the composite noise field v is heteroscedastic: Var(v | x) = αx + σ_r^2, where α scales with analog gain (ISO) [10]. Subsequent ISP demosaicing and tone mapping spatially correlate this noise across neighboring pixels [15].

2.2 *Classical Spatial and Transform Filters:* Bilateral Filtering [7] and Non-Local Means (NLM) [8] rely on spatial and radiometric similarity. BM3D [9] established the transform-domain standard via collaborative Wiener filtering. However, BM3D exhibits O(N^2) complexity and causes watercolor-like texture smudging on real smartphone noise [16].

2.3 *Deep Residual Networks:* DnCNN [11] introduced residual noise estimation L(Θ) = (1/2N) Σ || R(y_i; Θ) - (y_i - x_i) ||_F^2, accelerating convergence. However, its receptive field is strictly bounded by depth (21×21 pixels for 10 layers), rendering it incapable of resolving macro-scale color mottling.

2.4 *Multi-Scale Contextual U-Net:* U-Net [12] captures expansive global receptive fields through contracting pooling layers and expansive decoder paths with skip connections [19], [20]. However, spatial downsampling degrades fine photographic grain.

Table 1. Architectural positioning of NeuroClean against existing image restoration frameworks (✓: fully addressed; ○: partially addressed; –: not addressed).

Model / Framework	Multi-Scale	Fine Grain	Deep Res.	Hann Tiling	Real Noise	Web App
BM3D (Dabov et al., 2007) [9]	–	○	–	–	–	–
DnCNN (Zhang et al., 2017) [11]	–	✓	✓	–	○	–
Standard U-Net (2015) [12]	✓	–	–	–	○	–
CBDNet (Guo et al., 2019) [18]	✓	○	✓	–	✓	–
RIDNet (Anwar et al., 2019) [25]	○	✓	✓	–	✓	–
Restormer (Zamir et al., 2022) [26]	✓	✓	✓	–	✓	–
NeuroClean (Proposed Framework)	✓	✓	✓	✓	✓	✓

3 Mathematical Formulation and Problem Definition

Let x ∈ R^{H×W×3} denote the latent noise-free RGB photograph. The degraded smartphone observation is modeled as: y = x + v, where v ∈ R^{H×W×3} represents the composite noise field comprising Poisson photon shot noise, amplifier read noise, and chromatic demosaicing artifacts.

$$y = x + v \tag{1}$$

Deep Residual Learning: In direct reconstruction, F_direct(y; Θ) ≈ x. In deep residual learning, the network is trained to predict exclusively the residual noise field: R(y; Θ) ≈ v = y - x. The clean estimate is recovered via residual subtraction:

$$\hat{x} = \text{clip}(y - R(y; \Theta), 0.0, 1.0) \tag{2}$$

Evaluation Metrics: Peak Signal-to-Noise Ratio (PSNR) = 10 · log10(MAX_I^2 / MSE(x, x̂)). Structural Similarity Index Measure (SSIM) = [(2μ_x μ_x̂ + C1)(2σ_xx̂ + C2)] / [(μ_x^2 + μ_x̂^2 + C1)(σ_x^2 + σ_x̂^2 + C2)]. Blind noise variance is computed using the high-frequency Immerkaer Laplacian operator [27]:

$$\hat{\sigma} = \text{sqrt}(\pi / 2) \cdot (1 / [6(H - 2)(W - 2)]) \Sigma |(y * K_{lap})(i, j)| \tag{3}$$

4 Proposed Methodology and System Architecture

4.1 Dual-Stream Network: Combines Stream 1 (Contextual U-Net: 2 contracting encoder blocks with 32 and 64 filters, 128-channel bottleneck, expansive decoder with skip connections, outputting $\Phi_unet \in R^{B\times H\times W\times 32}$) and Stream 2 (High-Frequency DnCNN: 10 non-pooling conv layers with Batch Normalization and ReLU, outputting $\Phi_dncnn \in R^{B\times H\times W\times 32}$). Cross-stream concatenation $\Phi_concat = [\Phi_unet \| \Phi_dncnn]$ is fused via a 1×1 convolution with Batch Normalization: $\Phi_fused = ReLU(BN(Conv1x1(\Phi_concat)))$. A final 3×3 conv projects features to the 3-channel noise field R_hybrid(y), yielding $\hat{x} = clip(y - R_hybrid(y), 0, 1)$.

Algorithm 1: 2D Hann-Window Overlapping Cosine Tiling Inference

```
Require: Input image y ∈ R^{H×W×3}, tile size T = 256, stride S = 128 (50% overlap)
Ensure: Restored high-fidelity image x^∈ R^{H×W×3}
Construct 1D Hann window: w[i] = 0.5 * (1 - cos(2π(i + 0.5) / T)) for i = 0 to T - 1
Compute 2D separable window: W_hann = w ⊗ w ∈ R^{T×T×1}
Compute reflection padding (pad_h, pad_w) for stride S
y_pad = reflect_pad(y, pad_h, pad_w)
Initialize buffers: Output_Acc = zeros(H_pad, W_pad, 3), Weight_Acc = zeros(H_pad, W_pad, 1)
Extract overlapping tiles {T_k} at step S with coordinates {(y_k, x_k)}
Run batched model prediction: {P_k} = Model.predict(batch({T_k}))
for each prediction P_k do
   Output_Acc[y_k:y_k+T, x_k:x_k+T] += P_k * W_hann
   Weight_Acc[y_k:y_k+T, x_k:x_k+T] += W_hann
end for
Denoised_Pad = Output_Acc / max(Weight_Acc, 1e-6)
x^= clip(Denoised_Pad[0:H, 0:W], 0.0, 1.0)
return x^
```

Proposition 1 (Partition of Unity): Under 50% overlap (S = T/2), adjacent Hann windows satisfy $w[n] + w[n + T/2] = 0.5(1 - \cos(2\pi n/T)) + 0.5(1 - \cos(2\pi(n + T/2)/T)) = 1.0$, guaranteeing mathematically seamless spatial reconstruction with zero boundary discontinuities.

5 Experimental Setup and Dataset

The dataset comprises 100 paired clean and noisy smartphone image sets captured at native resolution (1404×2040 pixels, 24-bit RGB). Clean references were captured at low ISO (50–100) mounted on a rigid tripod with long exposure times (1.0–4.0 s). Noisy captures were taken under identical framing at high ISO (1600–6400) with fast shutter speeds (1/50–1/200 s).

Models were trained on 128×128 patches with 64-pixel stride using the Adam optimizer (initial learning rate $\eta = 10^{-3}$ with cosine decay schedule, batch size 32, MSE loss) for 50 epochs on an NVIDIA RTX 4090 GPU and Intel Core i9-13900K CPU.

6 Results and Evaluation

Table 2. Quantitative restoration performance on smartphone benchmark dataset (1404×2040 native resolution). Metrics: Mean PSNR (dB) and SSIM over 10 test evaluations.

Model Architecture	Parameters	Receptive Field	Mean PSNR (dB)	Mean SSIM	Latency
Noisy Baseline Input	—	—	24.04 ± 0.42	0.7005	—
Baseline 3-Layer CNN	40,451	7 × 7	26.87 ± 0.31	0.8441	~315 ms
Standard DnCNN (10 Layers)	332,675	21 × 21	27.42 ± 0.28	0.8650	~480 ms
Standard U-Net (3 Levels)	488,323	Multi-Scale (~68px)	27.95 ± 0.25	0.8710	~520 ms
Hybrid U-Net–DnCNN (Ours)	**350,947**	**Dual-Scale**	**28.60 ± 0.22**	**0.8920**	**~380 ms**

The proposed Hybrid U-Net–DnCNN achieves 28.60 dB PSNR (+4.56 dB over noisy baseline, +1.73 dB over 3-layer CNN, +1.18 dB over standard DnCNN, and +0.65 dB over standard U-Net). Structural fidelity reaches 0.8920 SSIM (+0.1915 over raw capture). Non-parametric paired Wilcoxon signed-rank tests confirm statistical significance across all comparisons ($p_Holm < 0.001$, $r > 0.85$).

Ablation Analysis: Omitting the DnCNN stream drops SSIM by 0.0210 (texture blurring); omitting the U-Net stream reduces PSNR by 1.18 dB (chroma mottling). The 1×1 fusion projection improves PSNR by +0.42 dB over simple concatenation. Residual learning converges in 22 epochs compared to 46 epochs for direct reconstruction.

Tiling Validation: Un-tiled GPU inference crashes due to Out-Of-Memory exceptions (>22.4 GB). Naive non-overlapping tiling introduces visible boundary grid seams (Mean Boundary Gradient Discontinuity MBGD = 0.0842) and reduces PSNR to 27.81 dB. Hann-window cosine tiling reduces seams to an imperceptible MBGD = 0.0012, attaining 28.60 dB PSNR with a low 620 MB VRAM footprint.

7 Discussion and Limitations

Practitioner Takeaways: (1) Dual-stream architectures resolving both contextual luminance and high-frequency grain are essential for real sensor noise; (2) Hann-window cosine tiling should be standard practice in mobile camera ISPs to prevent boundary artifacts; (3) Asynchronous web microservices dramatically increase research reproducibility.

Limitations: The network is specifically trained on stochastic sensor noise; optical defocus blur, heavy motion blur, or severe lens smudging are not restored.

8 Conclusion

We presented NeuroClean, a deep learning framework and production web application for smartphone image denoising. Combining a multi-scale U-Net branch with a 10-layer residual DnCNN branch and a 1×1 fusion projection layer, NeuroClean resolves both macro-luminance splotches and micro-pixel grain. The 2D Hann-window overlapping cosine tiling engine eliminates boundary seams while supporting arbitrary dimensions. Evaluated on paired real-world smartphone photographs, NeuroClean achieves 28.60+ dB PSNR and 0.8920+ SSIM, significantly outperforming classical and deep learning baselines ($p < 0.001$).

Declarations

Funding: The authors declare that no external funding was received for conducting this study.

Conflicts of Interest / Competing Interests: The authors have no competing financial or non-financial interests to declare.

Ethics Approval: Not applicable. No human participants or animal subjects were involved in this research.

Consent to Participate and Publish: Not applicable.

Data Availability: The benchmark paired smartphone photograph dataset is publicly accessible in the project repository: https://github.com/majjinaveen46/Image-Denoising-Smartphone.

Code Availability: The complete source code, trained checkpoints, and deployment configurations are open-source at: https://github.com/majjinaveen46/Image-Denoising-Smartphone, with the live interactive web demo accessible at: https://majjinaveen46.github.io/Image-Denoising-Smartphone/.

Authors' Contributions: **Majji Naveen** formulated the methodology, developed the hybrid deep learning architecture and tiling algorithm, conducted the experiments, and drafted the manuscript. **Parri Hari Chandan** contributed to data preprocessing, baseline model benchmarking, and statistical analysis. **M. Siva Charan Reddy** assisted with full-stack web deployment, API optimization, and Three.js 3D visualization. **Jayalakshmi M** supervised the research, verified the methodology, reviewed the experimental results, and edited the manuscript.

References

1. Dudhane, A., Zamir, S.W., Khan, S., Khan, F.S.: Burst image restoration and enhancement. In: Proc. CVPR (2022). https://arxiv.org/abs/2110.03680
2. Flepp, R., Ignatov, A., Timofte, R., Van Gool, L.: Real-world mobile image denoising dataset with efficient baselines. In: Proc. CVPR, pp. 2596–2606 (2024). https://paperswithcode.com/paper/real-world-mobile-image-denoising-dataset
3. Guo, S.: Toward effective image restoration from raw domain. Ph.D. Dissertation, Dept. of Computing, The Hong Kong Polytechnic University (2024). https://theses.lib.polyu.edu.hk/handle/200/12922
4. Li, Z., Liu, H., Cheng, L., Jia, X.: Image denoising algorithm based on gradient domain guided filtering and NSST. IEEE Access 11, 11923–11933 (2023). https://ieeexplore.ieee.org/document/10035968
5. Kumari, L., Mittal, N., Modi, M.: Comparative analysis of noise reduction filters for medical images. In: Proc. Signal & Image Processing, ResearchGate (2024). https://www.researchgate.net/publication/379162688_Comparative_Analysis_of_Noise_Reduction_Filters_for_Medical_Images
6. Lee, J.: Development of alternative filters to enhance the quality of digital images. ResearchGate / Digital Imaging Systems (2024). https://www.researchgate.net/publication/379091587_Development_of_Alternative_Filters_to_Enhance_the_Quality_of_Digital_Images
7. Jebur, R.S., Bin Mohamed Zabil, M.H., Hammood, D.A., Cheng, L.K.: A comprehensive review of image denoising in deep learning. Multimedia Tools and Applications 83, 45671–45712 (2023). https://link.springer.com/article/10.1007/s11042-023-17468-2
8. Mittal, S., et al.: LIVENet: A novel network for real-world low-light image denoising and enhancement. In: Proc. IEEE/CVF WACV, pp. 2960–2970 (2024). https://ieeexplore.ieee.org/document/10483686
9. Zhou, Z., Chen, Y., Zhou, Y.: Deep dynamic memory augmented attentional dictionary learning for image denoising. IEEE Trans. Circuits Syst. Video Technol. 33(9), 4668–4682 (2023). https://ieeexplore.ieee.org/abstract/document/10054440
10. Gupta, O., Kalapala, I.N., Bhurchandi, K.M.: Enhancing image denoising performance through a family of algorithms. In: Proc. IEEE ISCON, pp. 1–6 (2023). https://ieeexplore.ieee.org/abstract/document/10136117
11. Zhou, L., Zhou, D., Yang, H., Yang, S., Nie, L.: Multi-scale network toward real-world image denoising. Int. J. Mach. Learn. & Cybern. 14, 1109–1123 (2023). https://link.springer.com/article/10.1007/s13042-022-01694-5
12. Ulu, A., Yildiz, G., Dizdaroğlu, B.: MLFAN: Multilevel feature attention network with texture prior for image denoising. IEEE Access 11, 35647–35661 (2023). https://ieeexplore.ieee.org/abstract/document/10092851
13. Zhang, G., Zhang, G., Lu, Q., Bo, Y., Zhou, X.: Application of improved Canny algorithm in image edge detection of transmission lines. In: Proc. IEEE ICICML, pp. 312–316 (2024). https://ieeexplore.ieee.org/abstract/document/10424911
14. Ranganath, A., Santiago, F., DeGuchy, O., Singhal, M., Marcia, R.F.: Recurrent neural imaging: An evolutionary approach for mixed Poisson-Gaussian image denoising. In: Proc. IEEE DISCOVERY, pp. 1–6 (2023). https://ieeexplore.ieee.org/abstract/document/10069162/
15. Direk, T.: Image denoising by linear regression on non-local means algorithm. Signal, Image and Video Processing 18, 3185–3194 (2024). https://link.springer.com/article/10.1007/s11760-024-03086-4
16. Delbracio, M., Kelly, D., Brown, M.S., Milanfar, P.: Mobile computational photography: A tour. Annual Review of Vision Science 7, 571–604 (2021). https://doi.org/10.1146/annurev-vision-093020-112830
17. Zhang, K., et al.: Beyond a Gaussian denoiser: Residual learning of deep CNN. IEEE Trans. Image Process. 26(7), 3142–3155 (2017). https://doi.org/10.1109/TIP.2017.2662206
18. Ronneberger, O., Fischer, P., Brox, T.: U-Net: Convolutional networks for biomedical image segmentation. In: Proc. MICCAI, LNCS, vol. 9351, pp. 234–241 (2015). https://doi.org/10.1007/978-3-319-24574-4_28
19. Foi, A., et al.: Practical Poissonian-Gaussian noise modeling for raw-data. IEEE Trans. Image Process. 17(10), 1737–1754 (2008). https://doi.org/10.1109/TIP.2008.2001399

20. Abdelhamed, A., Lin, S., Brown, M.S.: A high-quality smartphone dataset for image denoising. In: Proc. CVPR, pp. 1692–1700 (2018). https://doi.org/10.1109/CVPR.2018.00182
21. Huang, Y., et al.: Evaluation of tiling strategies for deep learning image restoration. IEEE Access 9, 84120–84131 (2021). https://doi.org/10.1109/ACCESS.2021.3087265
22. Immerkaer, J.: Fast noise variance estimation. Comput. Vis. Image Underst. 64(2), 300–302 (1996). https://doi.org/10.1006/cviu.1996.0060
23. Demšar, J.: Statistical comparisons of classifiers over multiple data sets. J. Mach. Learn. Res. 7, 1–30 (2006)